

VENOUS THROMBOEMBOLISM

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Venous Thromboembolism (VTE)

- VTE results from **clot formation in the venous circulation** and is manifested as
 - Deep vein thrombosis (DVT) and
 - Pulmonary embolism (PE)
- It is often provoked by **prolonged immobility** and **vascular injury** and is most frequently seen in pts who have been **hospitalized** for a serious medical illness, trauma, or major surgery.

- **DVT** is a thrombus composed of cellular material (RBC and WBC, platelets) bound together with fibrin strands mainly in legs and arm.
- **PE** is a thrombus that arises from the systemic circulation and lodges in the pulmonary artery or one of its branches, causing complete or partial obstruction of pulmonary blood flow.

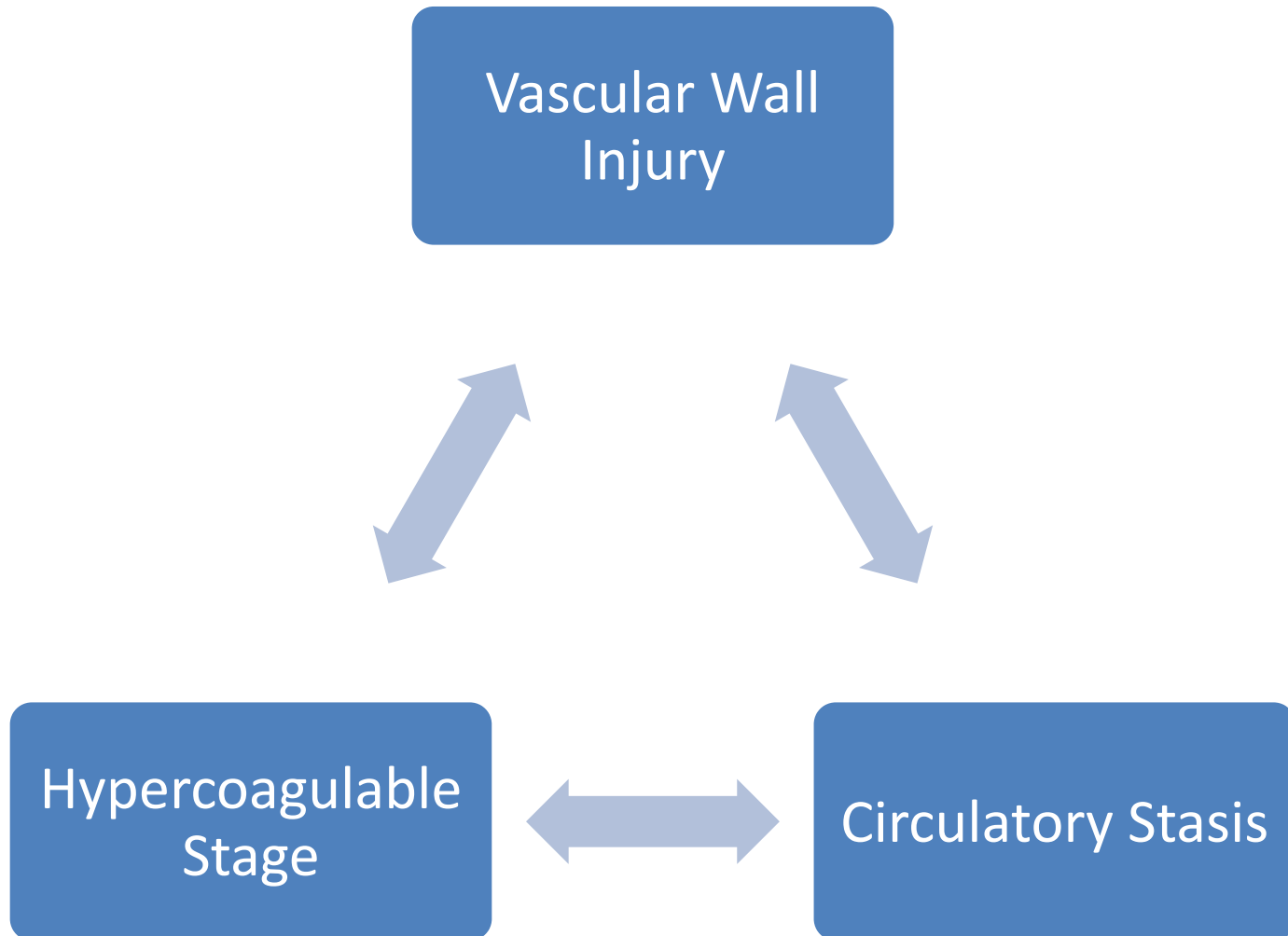
Epidemiology

- VTE annual incidence: 1 in 1000 cases
- VTE is an important cause of morbidity and mortality, particularly in hospitalized pts.
- PE is the cause of death or a major contributing factor in up to 16% of pts who die in the hospital.
- PE has 50% higher incidence in AfAm compared to Caucasians.
- The dx is often clinically **silent**, and the first manifestation may be **sudden death**.

Pathophysiology

- Coagulation cascade can be triggered through either the **intrinsic** or **extrinsic** pathways.
- **Three primary components (Virchow's triad)** play a role in the development of thrombus.
 - Venous stasis,
 - Vascular injury, and
 - Hypercoagulability

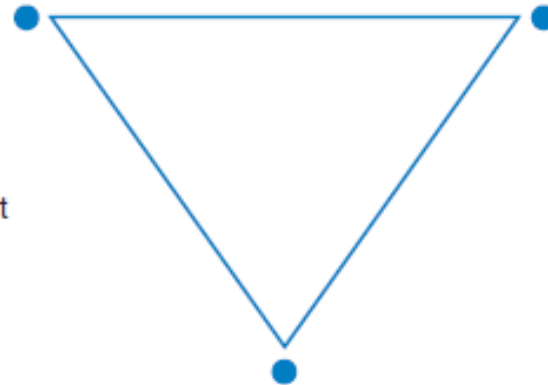
Virchow's Triad



Risk factors for thromboembolism

Abnormalities of Blood Flow

- Atrial fibrillation
- Bed rest/immobilization/paralysis
- Left ventricular dysfunction from:
 - ischemic or idiopathic cardiomyopathy, congestive heart failure, or myocardial infarction
- Venous obstruction from tumor, obesity, or pregnancy



Abnormalities of Surfaces in Contact With Blood

- Acute myocardial infarction
- Atherosclerosis
- Chemical irritation (potassium, hypertonic solutions, chemotherapy)
- Fractures
- Heart valve disease
- Heart valve replacement
- Indwelling catheters
- Previous DVT or PE
- Tumor invasion
- Vascular injury or trauma

Abnormalities of Clotting Components

- Antiphospholipid antibody syndrome (lupus anticoagulant; anticardiolipin antibody)
- Antithrombin deficiency
- Dysfibrinogenemia
- Estrogen therapy
- Factor V Leiden
- Homocysteinemia
- Malignancy
- Myeloproliferative disorder
- Polycythemia
- Pregnancy
- Protein C deficiency
- Protein S deficiency
- Prothrombin G20210A mutation
- Thrombocytosis

- **Venous stasis** is slowed blood flow in the deep veins of the legs resulting from
 - Damage to venous valves
 - Vessel obstruction
 - Prolonged periods of immobility or
 - Increased blood viscosity
- **Vascular injury** may result from
 - Major orthopedic surgery (**knee and hip replacement**),
 - Trauma (especially fractures of the pelvis, hip or leg), or
 - Indwelling venous catheters

- **Hypercoagulable states** include
 - Malignancy
 - Activated protein C resistance
 - Deficiency of protein C, protein S, or antithrombin
 - Factor VIII or XI excess
 - Antiphospholipid antibodies and
 - **Estrogens and selective estrogen receptor modulators** have been linked to venous thrombosis

Table 15-1 Inhibitors of Clotting Mechanisms

<i>Inhibitor</i>	<i>Target</i>
Antithrombin	Inhibits factors IIa, IXa, and Xa
Protein S	Cofactor for activation of protein C
Protein C	Inactivates factors Va and VIIIa
Tissue factor pathway inhibitor	Inhibits activity of factor VIIa
Plasminogen	Converted to plasmin via tissue plasminogen activator
Plasmin	Lyses fibrin into fibrin degradation products

TABLE 7–1. Risk Factors for Venous Thromboembolism

Risk Factor	Example
Age	Risk doubles with each decade after age 50
Prior history of VTE	Strongest known risk factor for DVT and PE
Venous stasis	Major medical illness (e.g., congestive heart failure) Major surgery (e.g., general anesthesia for greater than 30 minutes) Paralysis (e.g., due to stroke or spinal cord injury) Polycythemia vera Obesity Varicose veins
Vascular injury	Major orthopedic surgery (e.g., knee and hip replacement) Trauma (esp. fractures of the pelvis, hip, or leg) Indwelling venous catheters
Hypercoagulable states	Malignancy, diagnosed or occult Activated protein C resistance/factor V Leiden Prothrombin (20210A) gene mutation Protein C deficiency Protein S deficiency Antithrombin deficiency Factor VIII excess (greater than 90th percentile) Factor XI excess (greater than 90th percentile) Antiphospholipid antibodies Dysfibrinogenemia PAI-I excess Pregnancy/postpartum
Drug therapy	Estrogen-containing oral contraceptive pills Estrogen replacement therapy SERMs HIT

Risk Factors for Predicting Venous Thromboembolism (VTE) in Hospitalized Medical Patients (Padua Prediction Score)

Risk Factor	Points
Active cancer ^a	3
Previous VTE (with the exclusion of superficial vein thrombosis)	3
Reduced mobility ^b	3
Already known thrombophilic condition ^c	3
Recent (≤ 1 month) trauma and/or surgery	2
Elderly age (≥ 70 years)	1
Heart and/or respiratory failure	1
Acute myocardial infarction or ischemic stroke	1
Acute infection and/or rheumatologic disorder	1
Obesity (body mass index ≥ 30)	1
Ongoing hormonal treatment	1
Cumulative score of ≥ 4 points indicates high risk of VTE	

Clinical Presentation

DVT

- Pain
- Tenderness
- Unilateral leg swelling
- Warmth
- Erythema
- Palpable cord
- Asymptomatic

*DVT cannot be diagnosed or excluded
based upon presentation.

PE

- Chest pain, SOB
- Tachypnea, Tachycardia
- Pain (back, shoulder, upper abdominal)
- Syncope
- Hemoptysis
- New cardiac arrhythmia
- Palpitation
- Cough

DVT



Diagnosis

- Assessment of the pt's status should focus on the search for **risk factors**
 - Increased age -Major surgery
 - Previous VTE -Trauma
 - Malignancy -Hypercoagulable states
 - Drug therapy
- Signs & symptoms of DVT are nonspecific & **objective tests** are required to confirm or exclude the diagnosis.

- **Radiographic contrast** studies are the most accurate and reliable method for diagnosis of VTE.
 - **Contrast venography** allows visualization of the entire venous system in the lower extremity & abdomen.
 - **Pulmonary angiography** allows visualization of the pulmonary arteries.
- Ultrasonography, CT scans & ventilation-perfusion scan
- **D-dimer test** is a very sensitive marker of clot formation

Treatment

- **The objectives of treating VTE are to:**
 - Prevent the development of PE and the post-thrombotic syndrome,
 - Reduce morbidity & mortality
 - Minimize adverse effects and cost of treatment.

Heparin

- **UFH prevents** the growth and propagation of a formed thrombus & allows the pt's own thrombolytic system to degrade the clot.
- **Contraindications to heparin** include
 - Hypersensitivity to the drug, Active bleeding, Hemophilia,
 - Severe liver disease with elevated PT,
 - Severe thrombocytopenia, Malignant hypertension, and
 - Inability to meticulously supervise and monitor treatment
- **UFH must be given parenterally, preferably by the IV or SC**

- LMWHs are fragments of UFH that are heterogeneous mixtures of sulfated glycosaminoglycans with approximately $1/3^{\text{rd}}$ the molecular weight of UFH.
- Like UFH, the LMWHs enhance & accelerate the **activity of antithrombin &** prevent the growth & propagation of formed thrombi.
- The peak anticoagulant effect is seen in **3 to 5 hrs** after SC dosing.

- ✓ **Enoxaparin (Lovenox)** 1 mg/kg every 12 hrs or 1.5 mg/kg every 24 hrs
- ✓ **Dalteparin (Fragmin)** 100 u/kg every 12 hrs or 200 u/kg every 24 hrs
- ✓ **Tinzaparin (Innohep)** 175 u/kg every 24 hrs

- **Fondaparinux sodium (Arixtra)** is a selective inhibitor of **factor Xa**.
- It is approved for **prevention of VTE in pts undergoing orthopedic** (hip fracture, hip & knee replacement) surgery and **for treatment of VTE and PE**.
 - **VTE prevention**, the dose is **2.5 mg SC QD** starting 6-8hrs after surgery.
 - **Treatment of DVT and PE**: the usual dose is **7.5 mg SC QD**

- These agents interact directly with **thrombin**.
- They are capable of inhibiting both circulating and clot-bound thrombin, which is a potential advantage over UFH and the LMWHs.
- Are indicated for anticoagulation in pts with HIT
 - Lepirudin (Refludan)
 - Bivalirudin (Angiomax)
 - Desirudin (Iprivask)

- **Warfarin** inhibits cyclic interconversion of vitamin K in the liver and reduced vitamin K.
- For pts with acute VTE, heparin and warfarin therapy should be overlapped for at **least 4 - 5 days**, regardless of whether the target INR has been achieved earlier.
- The UFH or LMWH can then be discontinued once the INR is within the desired range for 2 consecutive days
- Monitor **INR** (target: 2-3 DVT/PE)

- Thrombolytic agents degrade the fibrin matrix
- Three thrombolytic agents and regimens are available for treatment of DVT and/or PE in selected pts:
 - ✓ **Streptokinase (Streptase):** 250,000 units IV over 30 min followed by a continuous IV infusion of 100,000 units/hr for 24 hrs (PE) or 24 to 72 hrs (DVT).
 - ✓ **Urokinase (Abbokinase):** For PE, 4,400 IU/kg IV over 10 min followed by 4,400 IU/kg/hr for 12 to 24 hrs.
 - ✓ **Alteplase (Activase):** For PE, 100 mg by IV infusion over 2 hrs

DVT Prophylaxis Therapy

	Heparin	Fragmin	Lovenox	Arixtra	Warfarin
Hip Replacement	7500 u SC BID or TID	2500 u SC QD or 5000u SC QD	30mg SC Q12H or 40mg SC QD	2.5mg SC QD x 5-9 days	2-5mg/day IV (administer as a slow bolus injection) INR: 2-3 Initial dosing individualized Start 2-5mg/day for two days or 5- 10mg/day for 1-2 days and adjust dose to INR response
Knee Replacement	5000 units SC BID or TID		30mg SC Q12H	2.5mg SC QD x 5-9 days	
Severely restricted mobility during acute illness	5000 units SC BID or TID	5000 u SC QD			
Abdominal Surgery	5000 units SC BID or TID	2500 units SC QD	40mg SC QD	2.5mg SC QD x 5-9 days	

Special Populations

	UFH	Lovenox	Arixtra
Renal Impairment	Preferred	Must Adjust dose for CrCl<30ml/min	Contraindicated in CrCl <30ml/min
Obesity	Preferred	Studied up to 196kg	Studied up to 226kg
Low Weight Patients/Frailty	N/A	Precaution in men <57kg and women <45kg	Contraindicated for prophylaxis <50kg for orthopedic /abdominal surgeries
Previous HIT	Contraindicated	Caution	No Mention
Pregnancy	Category C	Category B	Category B

Monitoring Parameters

	UFH	LMWH	Fondaparinux (Arixtra)	Warfarin
Complete Blood Counts	YES	YES	YES	YES
aPTT	YES	N/A	N/A	
Anti-Xa levels Obesity/pregnancy	N/A	YES	YES	N/A
Serum Creatinine	N/A	YES	YES	N/A
INR				2-3 days after initial dose

	Antidote/Dosing
UFH	Protamine Sulfate Binds 100% -1mg/100 units of heparin, cover for the last 3 hours of an IV infusion of heparin
LMWH	Protamine Sulfate Binds 60% -1mg Protamine sulfate per 1mg of enoxaparin -A second dose of 0.5mg proatmine sulfate per 100 anti-Xa units should be administered if bleeding continues.
Arixtra	Novoseven -90 mcg/kg intravenous bolus

Warfarin Reversal

Supratherapeutic range	Recommendation
INRs above therapeutic range but <5 with no significant bleeding	Lower dose or omit a dose, monitor more frequently, and resume therapy at appropriately adjusted dose when INR is at therapeutic level
INRs of ≥ 5 but <9 and no significant bleeding	Omit the next one or two doses, monitor more frequently and resume therapy at an appropriately adjusted dose when INR is at a therapeutic level. Alternately omit a dose and administer vitamin K orally if patient is at increased risk of bleeding
INRs ≥ 9 and no significant bleeding	Hold warfarin therapy and administer a higher dose of vitamin K (2.5 to 5mg) orally

Clinical Model/Modified Wells Criteria for Evaluating the Pretest Probability of Deep Vein Thrombosis (DVT)^a

Clinical Characteristic	Score
Active cancer (cancer treatment within previous 6 months or currently on palliative treatment)	+1
Paralysis, paresis, or recent plaster immobilization of the lower extremities	+1
Recently bedridden for 3 days or more, or major surgery within the previous 12 weeks requiring general or regional anesthesia	+1
Localized tenderness along the distribution of the deep venous system	+1
Entire leg swollen	+1
Calf swelling at least 3 cm larger than that on the asymptomatic side (measured 10 cm below tibial tuberosity)	+1
Pitting edema confined to the symptomatic leg	+1
Collateral superficial veins (nonvaricose)	+1
Previously documented DVT	+1
Alternative diagnosis at least as likely as DVT	-2